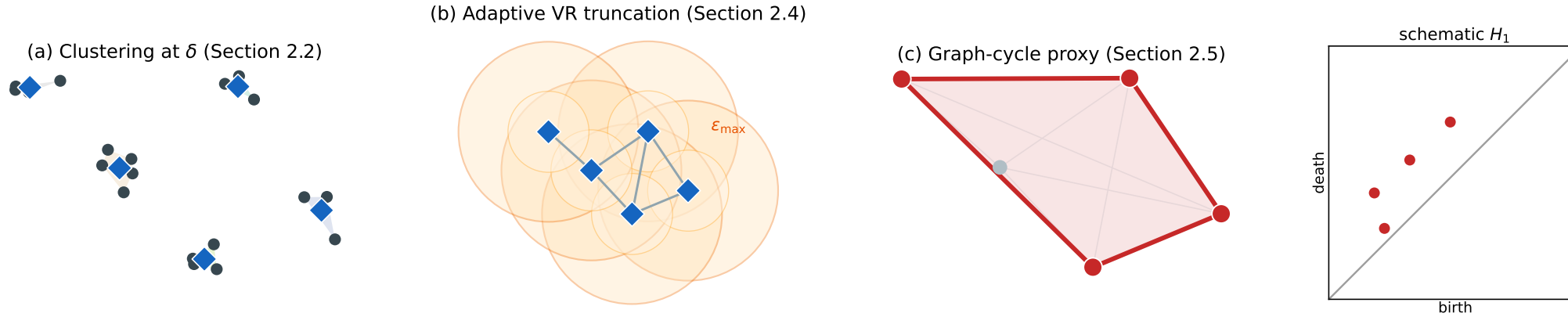
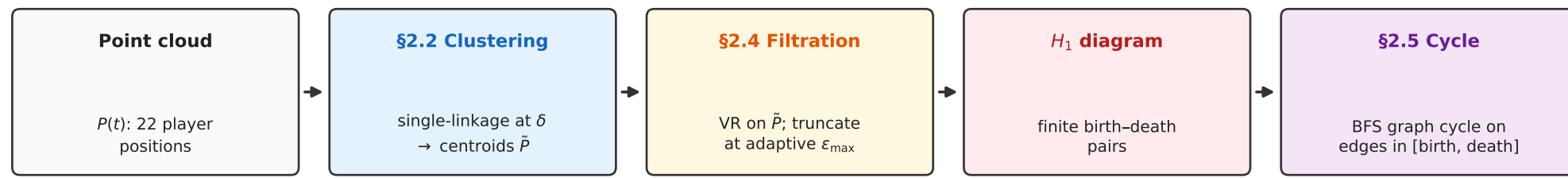


Multi-scale analysis pipeline



Homology is computed on cluster centroids $\tilde{P}(t)$, not on the raw 22-player cloud.

$\varepsilon_{\max} = \max(P_{75}\{d(\tilde{c}_i, \tilde{c}_j)\}, \max(5, 2\delta))$ truncates the Vietoris--Rips parameter (equation (1)); it is not a new filtration. The closed cycle is a geometric proxy for a persistence pair. Figure 2 shows the same cycle step on real tracking data.